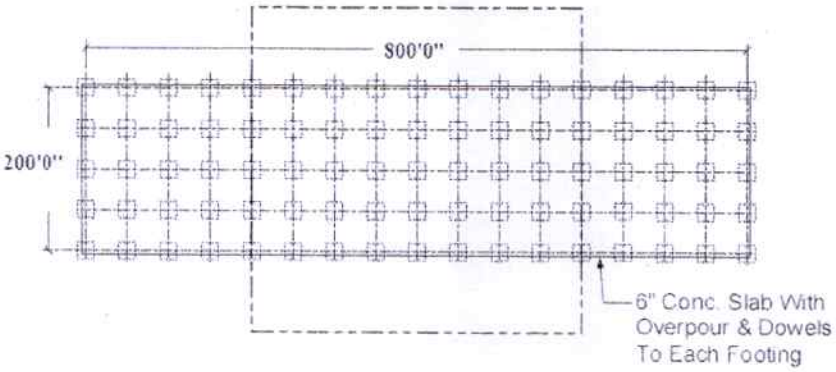
Roll No. | | | | | | | |

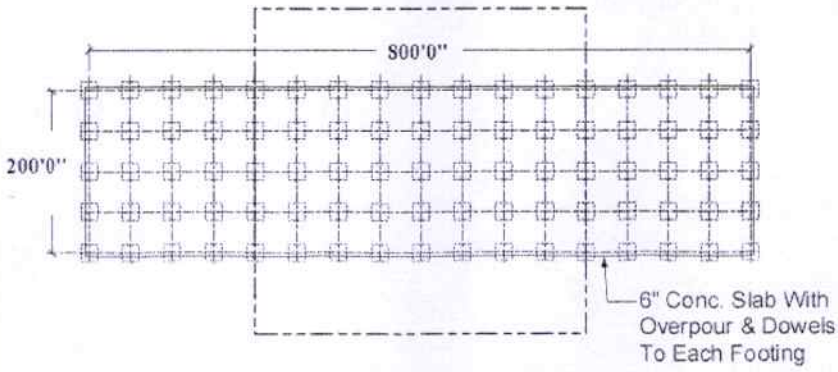
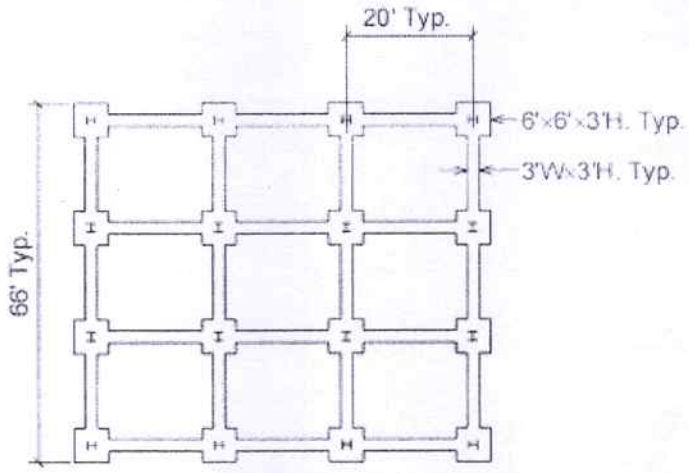
ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: M. Tech Structural Engineering Semester: III
Name of the Paper: Soil-Structure Interaction
Paper Code: MSE315
Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20marks)**	
(a)	Discuss the following SSI modeling approach (i) Substructure approach (ii) Direct Analysis Approach	
(b)	Discuss the following with illustration: Free-field motion, Foundation Input Motion, Kinematic Interaction, Inertial Interaction	**CO1**
(c)	Discuss in detail with illustration: How the building behaves and in the forces used in their design. Examples include: (i) Large building footprint, (ii) Substantial foundation embedment.	
Q2	**(20 marks)**	
(a)	Discuss how Large Building Footprint Reduces Design Forces?	
(b)	Give examples of "Building Base Configuration" with diagram showing different degrees of connectivity.	**CO2**
(c)	Discuss How Substantial Foundation Embedment Reduces Design Forces?	
Q3	**(20 marks)**	
(a)	Determine the fundamental time period of the structure of height 36 ft for which the value of the exponent is 0.75 and the coefficient is 0.020 according to ASCE/SEI 7-16.	**CO3**
(b)	With reference to the following diagram, calculate the Response spectral modification factor, if the effective foundation size is less than 260 ft and the minimum period allowed to be considered is 0.2 s.  	
(c)	With reference to the following diagram, calculate the B_{bsa} factor if the effective	

	foundation size is less than 260 ft and the minimum period allowed to be considered is 0.2 s.  	
Q4	**(20 marks)**	**CO4**
(a)	Highlight minimum 3 limitations to using base slab averaging effects in ASCE/SEI 7-16	
(b)	What do you understand by BASE SLAB AVERAGING. Discuss in detail	
(c)	Define the following terms related to base slab averaging • Equivalent foundation size • Effective Foundation size	
Q5	**(20 marks)**	**CO5**
(a)	Discuss the "embedment effect" of the foundation and list down the requirements to deal with the same for SSI effect.	
(b)	For the given building plan, calculate the ratio of footing area to the foot print area  	
(c)	Discuss the limitations and issues of using embedment effect as per ASCE/SEI 7-16.	